

Data Analysis
Course Project Report on

“GLOBAL CO₂ EMISSIONS ANALYSIS”

*SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF*

**BACHELOR OF TECHNOLOGY
IN**

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**VISHWAKARMA INSTITUTE OF TECHNOLOGY
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**DEPARTMENT OF ENGINEERING SCIENCE
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**BANSILAL RAMNATH AGARWAL CHARITABLE TRUST'S
VISHWAKARMA INSTITUTE OF TECHNOLOGY**

(An Autonomous Institute affiliated to Savitribai Phule Pune University)

2025 - 2026

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CERTIFICATE

This is to certify that the Course Project titled **“Global CO₂ Emissions Analysis”** submitted by **Yash Suvarnkar(1251130298), Swarali Salunkhe(1251130207), Swayam Chhatre(1251130196), Tanishq Tadas(1251130046)** is in partial fulfillment for the award of Degree of Bachelor of Technology in **Information Technology(IT)** of Vishwakarma Institute of Technology, Savitribai Phule Pune University. This project report is a record of bonafide work carried out by him under my guidance during the academic year 2025-26.

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Place: VIT, Pune
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Project Area : Data Science and Visual Analytics

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SIGNATURE OF INTERNAL GUIDE

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INTRODUCTION

In recent years, climate change and global warming have shifted from distant environmental warnings into immediate, undeniable global crises. Extreme weather events, rising sea levels, and unpredictable climate patterns are actively reshaping our world. At the absolute centre of this ecological disruption is the rapid increase of greenhouse gases in our atmosphere, primarily driven by carbon dioxide (CO₂) emissions. Ever since the industrial revolution, human activities—such as burning fossil fuels for electricity, running massive manufacturing plants, and expanding global transportation networks—have pumped billions of tons of carbon into the air. Understanding the exact trends, spikes, and drops in these emissions is no longer just a task for climate scientists; it has become a critical data-driven challenge for software engineers and data analysts who need to build tools to track and visualize this damage in real-time.

Our course project focuses on exploring and analysing global CO₂ emissions data, with a highly specific analytical lens placed on the transition period between the years 2018 and 2020. This particular timeline is incredibly significant from a historical and statistical standpoint. The year 2020 marked the global onset of the COVID-19 pandemic, an unprecedented event that forced governments worldwide to implement strict lockdowns, halt international and domestic travel, shut down commercial spaces, and temporarily suspend heavy industrial operations.

We wanted to see how these massive real-world events translate into raw data. By engineering an interactive data dashboard, our objective is to look past overwhelming spreadsheets and uncover how these socio-economic halts impacted carbon emissions across different continents, financial income groups, and individual countries. Through this data analysis project, we aim to prove that environmental tracking can be made accessible, clear, and highly visual, allowing anyone to see how human economic activity directly correlates with the carbon footprint left on our planet.

LITERATURE SURVEY

To build a strong foundation for our project, our team began by researching how carbon emissions and environmental data are currently tracked, managed, and presented across the tech industry. In the modern data landscape, there is no shortage of raw climate data. Massive global organizations continuously collect metrics on greenhouse gases. However, we quickly noticed a recurring problem: the data is usually trapped in giant, boring spreadsheets or locked behind highly technical academic interfaces that the average person or policymaker cannot easily interpret or interact with.

During our literature survey, we analysed some of the most prominent existing platforms in this domain to see what they do well and where they fall short:

1. Our World in Data (CO₂ and Greenhouse Gas Emissions Explorer):

This is undoubtedly one of the most comprehensive public repositories for climate data. It allows users to view country-level emissions stretching back over a century and create custom line graphs. However, from a user-experience (UX) perspective, it can feel incredibly cluttered. The platform requires navigating through multiple complex menus, dropdowns, and separate tabs. If a user wants a quick, consolidated, single-page snapshot of multi-dimensional data, such as looking at geographic locations alongside economic income brackets simultaneously, it becomes very tedious to configure.

2. The World Bank Data Portal:

The World Bank provides highly accurate datasets tracking carbon emissions per capita alongside development indicators. While their data is incredibly reliable, their presentation tools are largely static. Users are often forced to download massive CSV or Excel files and manually filter through thousands of rows of data to find meaningful insights. It completely lacks a modern, real-time interactive dashboard feel.

3. International Energy Agency (IEA) Data Browsers:

The IEA offers brilliant report-style dashboards focusing on energy-related emissions. However, their platforms are designed strictly for corporate energy analysts and climate scientists. The terminology used is highly technical, and the interfaces do not provide a straightforward, localized dashboard experience optimized for comparing specific, pivotal fiscal periods, like the immediate pre-pandemic year (2018) versus the peak pandemic year (2020).

Our project fills this exact gap by taking raw, unorganized global data and compiling it into a unified, single-page interactive dashboard application using Python. Instead of forcing a user to click through multiple pages or scroll through endless tables, we bring together key metrics—like overall global averages, regional variations, and economic class impacts—into a single, clean, dark-themed visual interface built for instant understanding.

METHODOLOGY

To transform raw environmental data into a fully functioning interactive dashboard, our group followed a structured, end-to-end data analysis workflow. This technical pipeline consisted of four main stages: data collection, rigorous preprocessing, data aggregation, and front-end dashboard development.

Data Collection Method & Source

- **Source of Data:** The dataset used for this analysis was sourced from open-source environmental tracking repositories and curated for academic data analysis. It covers detailed carbon emission records for over 180 countries.
- **Dataset Fields and Structure:** The raw data came in a structured tabular format (.csv). The core columns contained within our data file include:
 - **country code:** A unique three-letter ISO identifier for each country (e.g., USA, IND, AFG).
 - **country name:** The full official name of the nation.
 - **co2 2018:** The metric tons of \$CO_2\$ emissions per capita recorded in 2018.
 - **co2 2020:** The metric tons of \$CO_2\$ emissions per capita recorded in 2020.
 - **change 2018 to 2020:** The absolute numerical value change between the two target years.
 - **pct change 2018 to 2020:** The percentage increase or decrease in emissions.
 - **income group:** The economic classification of the country (e.g., High Income, Upper Middle Income, Lower Middle Income, Low Income).
 - **continent:** The geographic region where the country is located (e.g., Asia, Europe, Africa, North America, South America, Oceania).
 - **emission level 2020:** A categorical classification labelling countries from "Very Low" to "Very High" based on their carbon footprint intensity.

Tools and Technologies Used

As first-year IT students, we chose an entirely Python-based data stack because of its incredible efficiency, massive library support, and clean syntax:

- **Python:** The fundamental programming language used to write our processing logic and visualization scripts.
- **Pandas:** The industry-standard library used for backend data manipulation. We used Pandas to load the CSV, clean up any missing records, filter specific country groups, and perform quick math calculations.
- **Plotly & Plotly Subplots:** A powerful graphing library used to create interactive, web-ready charts. Unlike basic libraries like Matplotlib which create static images, Plotly allowed us to build charts that users can actually zoom into, hover over, and click.
- **Jupyter Notebook (.ipynb):** Our core development environment where we could write chunks of code, run them instantly, and see our visual outputs step-by-step.

Step-by-Step Implementation Pipeline

[Raw CSV Data] -> [Pandas Cleaning & Aggregation] -> [Plotly Subplots Grid Engine] -> [Interactive Dark Dashboard]

1. **Data Loading and Data Cleansing:** We began by using Pandas to read carbon_emission.csv. We scanned the dataset for any anomalies or missing values (NaN) and made sure that numerical columns were correctly formatted as floating-point numbers so our calculations wouldn't crash.
2. **Data Aggregation and Grouping:** To extract higher-level insights, we used the `..groupby()` function in Pandas. We grouped the countries by continent and income group to compute collective averages and total global sums, turning raw rows into meaningful group metrics.

3. **Dashboard Layout Architecture:** Using Plotly's *make_subplots* utility, we designed a master 2 x 2 grid canvas layout. We reserved the absolute top section of the canvas for broad KPI indicators, and mapped four distinct charts to specific row and column coordinates.

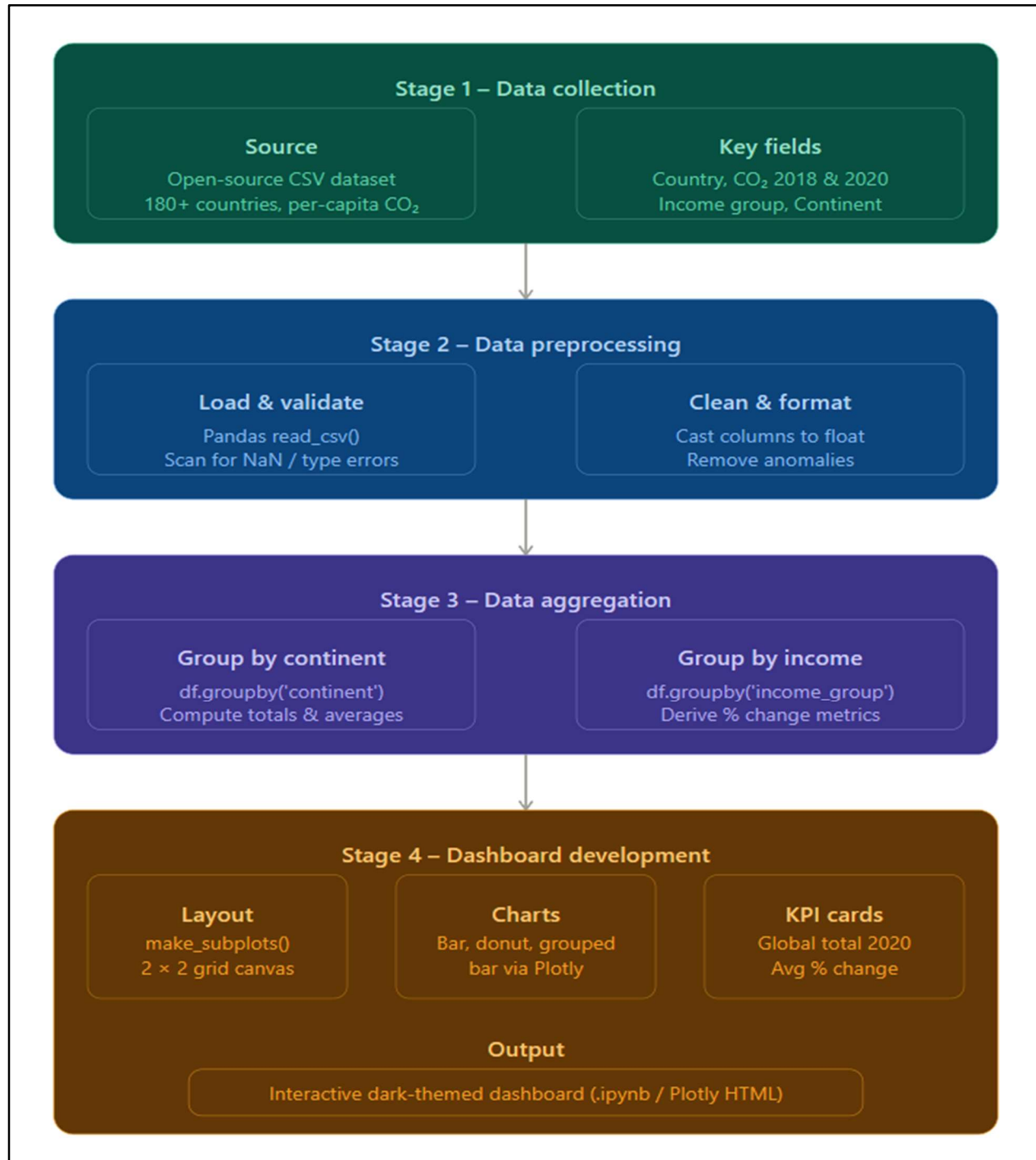


Fig. 1: Flowchart displaying the Methodology of the Project

FEATURES PROVIDED BY AN TOOL/APPLICATION

Our dashboard isn't just a collection of static graphs; it is designed as a complete, user-friendly interactive analytics tool. We engineered several key features to make sure the application is engaging and simple to navigate:

- **Real-Time Key Performance Indicators (KPI Cards):** Located prominently at the very top of the application dashboard, these big metric cards act as an executive summary. The moment a user launches the tool, they instantly see the massive macro-level figures—such as the total calculated global emissions and the global average percentage drop—giving immediate context before looking at individual graphs.
- **Dynamic Hover Interactivity:** Every single chart inside the dashboard is equipped with interactive tooltips. Built using Plotly's responsive rendering engine, users can hover their mouse cursor over any individual bar, slice, or segment to instantly view precise data points, country names, exact metric tons, and percentage values without cluttering the screen with text.
- **Multi-Dimensional Lens Analysis:** The tool allows users to break down and analyse the global carbon footprint from three completely different analytical angles within a single screen view:
 - *The Economic Lens:* Shows the clear connection between a nation's wealth and its pollution levels.
 - *The Geographic Lens:* Tracks absolute changes across physical world continents.
 - *The Intensity Classification Lens:* Groups countries into easy-to-understand labels based on their exact emission impact.
- **Synchronized Colour Mapping:** We carefully mapped specific, high-contrast colours to different datasets. This consistent design language ensures that a user can instantly make cross-chart comparisons without getting confused by shifting colour meanings.

SCREENSHOT OF DASHBOARD

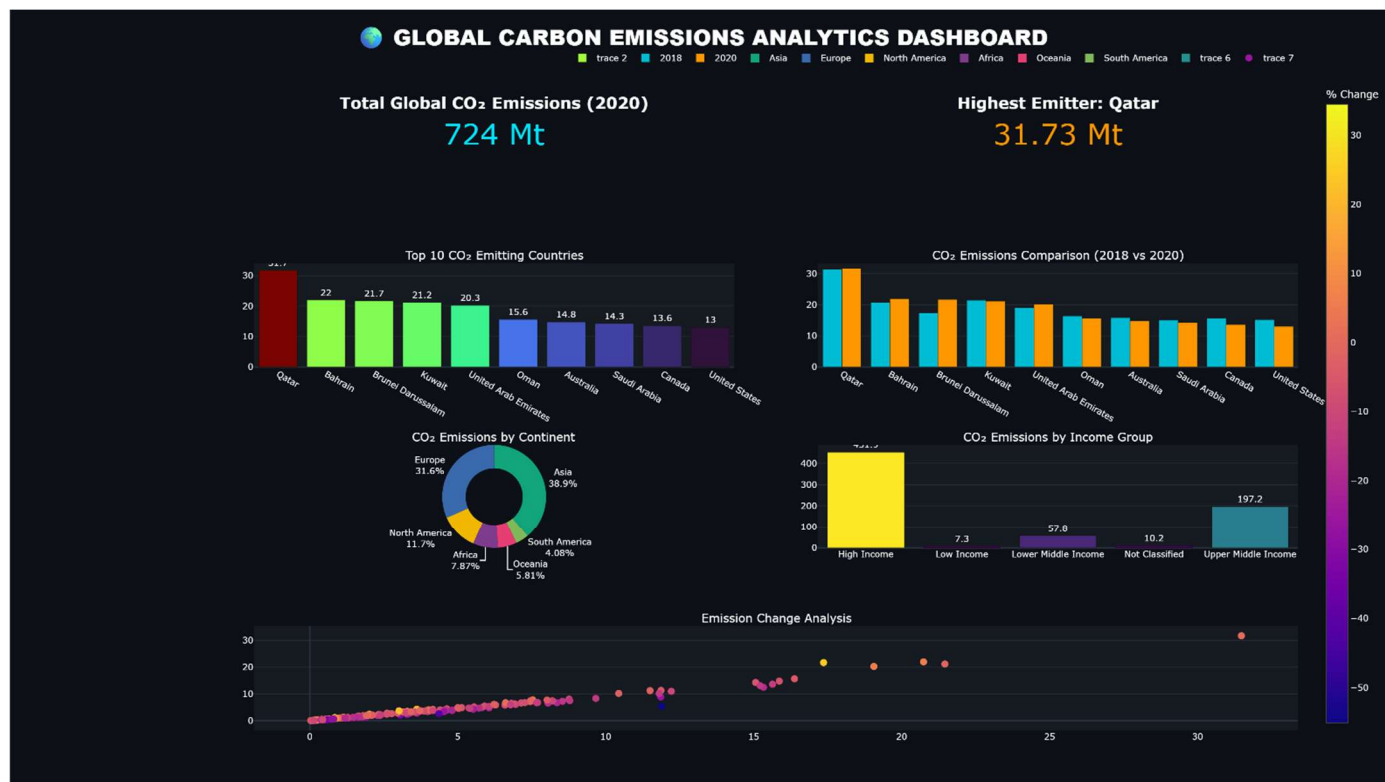


Fig. 2: The final Global CO₂ Emissions Analysis Dashboard

The screenshot displays our fully rendered, dark-themed Global \$CO_2\$ Emissions Analysis Dashboard. The entire user interface is structured cleanly into distinct visual zones, starting with the two major KPI Metric Indicators spanning across the top of the viewport:

- **Top Left Indicator (Total Global Emissions 2020):** This metric card calculates the collective per capita sum of emissions across the tracked nations, displaying a bold value of 724.36 Mt styled in an eye-catching electric cyan colour.
- **Top Right Indicator (Average Global % Change):** This card displays a value of -2.94% in vivid neon orange. This is a massive finding because it numerically confirms that, on a global scale, carbon emissions experienced a noticeable drop during the pandemic window due to halted human activities.

Directly below the top KPI section, the dashboard opens up into a balanced four-quadrant analytical visualization layout:

1. **Top-Left Bar Chart (Average CO₂ Emissions by Income Group):** This horizontal bar chart reveals an incredibly stark economic divide. High-income countries contribute vastly more to global emissions, averaging 5.42 metric tons per capita. In complete contrast, Low-income countries sit at a microscopic average of just 0.23 metric tons. This graph visually emphasizes that industrialization and national wealth are directly tied to environmental degradation.
2. **Top-Right Donut Chart (Distribution of Emission Levels 2020):** This chart categorizes all global nations by their emission intensity. It shows that while a massive portion of the world falls under the safer "Very Low (<1 t)" or "Low (1-3 t)" brackets, there is still a substantial, dangerous percentage of nations locked into the "High" and "Very High (>10 t)" brackets, indicating that carbon output is heavily concentrated in specific regions.

3. **Bottom-Left Grouped Bar Chart (Total Emissions by Continent - 2018 vs 2020):** This chart provides a direct comparison of emissions before and during the pandemic across six continents. By placing the 2018 bars (blue) right next to the 2020 bars (orange), you can see visible drops in North America, Europe, and Asia. This serves as empirical proof that lockdowns and reduced travel directly helped in temporarily flattening the global carbon emission curve.
4. **Bottom-Right Bar Chart (Average % Change in Emissions by Continent):** This final visualization captures the exact rate of change. While major industrial continents like Europe and North America show a significant negative percentage change (emissions dropping due to severe industrial lockdowns), continents like Africa and Oceania show a different trend. This helps us understand that economic recovery, industrial reliance, and pandemic lockdown policies varied wildly across different parts of the world.

CONCLUSION

Advantages of the Project

- **Highly Engaging UI/UX:** The dark-themed, sleek dashboard interface is incredibly appealing and makes complex environmental statistics vastly easier to read and interpret compared to traditional text-heavy academic spreadsheets.
- **Perfect Capture of Historical Anomalies:** Our application beautifully maps out the unique, real-world environmental impact of the 2020 COVID-19 lockdowns, acting as a valuable historical and educational data tool.
- **Holistic, Combined Analysis:** By merging geographic data alongside financial income classifications, the dashboard tells a complete, multi-layered story of exactly *who* is polluting the planet and *why*.

Shortcomings / Limitations of the Current Tool

- **Completely Static Data Source:** The backend relies entirely on a local, hardcoded carbon_emission.csv file. The dashboard cannot automatically update itself with live data from the web.
- **Highly Narrow Time Frame:** The analysis is tightly restricted to the years 2018 and 2020. Because of this, it fails to display long-term historical climbs (e.g., from the 1990s) or the inevitable post-pandemic carbon bounce-back seen in recent years.
- **Lack of Advanced Analytics:** The tool is strictly descriptive. It shows what happened in the past but does not feature any predictive algorithms to forecast where emissions are heading next.

FUTURE SCOPE

While our current dashboard is fully functional, there are several exciting ways we can scale and expand this application in the future using advanced software engineering practices:

- **Live API Integration & Automation:** We can replace our static CSV file by integrating live web APIs from organizations like the World Bank or Our World in Data. This would allow the dashboard to fetch and render the latest global emission figures automatically as soon as they are published.
- **Machine Learning Predictive Analytics:** By importing Python libraries like Scikit-Learn, we could build a predictive forecasting module. This feature would analyse historical trends and generate machine-learning-driven projections showing estimated carbon footprints for the next decade ().
- **Geographical Choropleth Mapping:** We plan to integrate interactive 3D world maps. By utilizing Plotly's geographic map features, we can make entire countries light up in different colour gradients based on their emission intensity, making spatial tracking much more immersive.
- **Granular Industrial Sector Breakdowns:** We want to expand our data layers to look beyond country names and isolate specific sectors, such as cutting data down by energy production, automotive transportation, agricultural farming, and tech data centres. This would tell us exactly which industries inside a nation are driving the emissions up.

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